

MVLU COLLEGE
A.I. Practical No.5

Aim:-Implement the SVM algorithm for binary classification.,
Train an SVM model using the IPL Most Runs 2020 dataset and optimize its parameters.,
Evaluate the performance of the SVM model on test data and analyze the results.

Code:-

```
Practical Spy - D:\Hari\AI\Practical Spy (3.13.5)
File Edit Format Run Options Window Help

import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
print("Hariprasad Vishwakarma T127")
df = pd.read_csv(r"F:\Downloads\Most Runs - 2020.csv")
print("Dataset loaded successfully!")
print("\nOriginal Dataset Shape:")
print(df.shape)

df = df.drop_duplicates()
print("\nDataset Shape After Removing Duplicates:")
print(df.shape)

if "HS" in df.columns:
    df["HS"] = df["HS"].astype(str).str.replace("+", "", regex=False)
    df["HS"] = pd.to_numeric(df["HS"], errors="coerce")

if "Player" in df.columns:
    df = df.drop("Player", axis=1)

if "POS" in df.columns:
    df = df.drop("POS", axis=1)

for column in df.columns:
    df[column] = pd.to_numeric(df[column], errors="coerce")

df = df.dropna()
print("\nMissing values removed successfully!")

median_runs = df["Runs"].median()

df["Performance"] = df["Runs"].apply(
    lambda x: 1 if x >= median_runs else 0
)

print("\nMedian Runs:")
print(median_runs)

print(median_runs)

print("\nTarget Classes:")
print(df["Performance"].value_counts())

X = df.drop(["Runs", "Performance"], axis=1)
y = df["Performance"]

print("\nNumber of Features:")
print(X.shape[1])

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)

print("\nTraining Samples:")
print(len(X_train))

print("\nTesting Samples:")
print(len(X_test))

pipeline = Pipeline([
    ("scaler", StandardScaler()),
    ("svm", SVC())
])

parameters = {
    "svm__C": [0.1, 1, 10, 100],
    "svm__kernel": ["linear", "rbf"]
}

grid_search = GridSearchCV(
    pipeline,
    parameters,
    cv=3,
    scoring="accuracy",
    n_jobs=1
)

print("\n=====")
print("Optimizing SVM Parameters...")
print("Please wait...")
```

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```
Practical 5.py - D:\Hari\AI\Practical 5.py (3.13.5)
File Edit Format Run Options Window Help
print("\n\n=====")
print("Optimizing SVM Parameters...")
print("Please wait...")
print("=====")

grid_search.fit(X_train, y_train)

print("\nBest Parameters:")
print(grid_search.best_params_)

print("\nBest Cross-Validation Accuracy:")
print(round(grid_search.best_score_ * 100, 2), "%")

best_model = grid_search.best_estimator_

y_pred = best_model.predict(X_test)

print("\nPrediction completed successfully!")

accuracy = accuracy_score(y_test, y_pred)

print("\nTest Accuracy:")
print(round(accuracy * 100, 2), "%")

print("\nClassification Report:")
print(classification_report(
    y_test,
    y_pred,
    target_names=["Low Performance", "High Performance"],
    zero_division=0
))

cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)

plt.figure(figsize=(7, 6))
plt.imshow(cm)
plt.title("SVM Confusion Matrix")
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")

plt.xticks(
    [0, 1],
    ["Low Performance", "High Performance"]
)

plt.yticks(
    [0, 1],
    ["Low Performance", "High Performance"]
)

for i in range(2):
    for j in range(2):
        plt.text(
            j,
            i,
            str(cm[i, j]),
            ha="center",
            va="center"
        )

plt.colorbar()
plt.tight_layout()
plt.show()

print("\n\n=====")
print("SVM PRACTICAL COMPLETED")
print("=====")

print("\nFinal Accuracy:")
print(round(accuracy * 100, 2), "%")

print("\nBest Parameters:")
print(grid_search.best_params_)

print("=====")
```

Output:-

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```
"IDLE Shell 3.13.5"
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: D:/Hari/A.I/Practical 5.py =====
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Dataset loaded successfully!

Original Dataset Shape:
(133, 14)

Dataset Shape After Removing Duplicates:
(133, 14)

Missing values removed successfully!

Median Runs:
59.0

Target Classes:
Performance
1 67
0 66
Name: count, dtype: int64

Number of Features:
11

Training Samples:
106

Testing Samples:
27

=====
Optimizing SVM Parameters...
Please wait...
=====

Best Parameters:
({'svm__C': 10, 'svm__kernel': 'rbf'})

Best Cross-Validation Accuracy:
98.12 %

=====
Name: count, dtype: int64

Number of Features:
11

Training Samples:
106

Testing Samples:
27

=====
Optimizing SVM Parameters...
Please wait...
=====

Best Parameters:
({'svm__C': 10, 'svm__kernel': 'rbf'})

Best Cross-Validation Accuracy:
98.12 %

Prediction completed successfully!

Test Accuracy:
100.0 %

Classification Report:
      precision    recall  f1-score   support

Low Performance      1.00      1.00      1.00        13
High Performance      1.00      1.00      1.00        14

   accuracy      1.00      1.00      1.00        27
  macro avg      1.00      1.00      1.00        27
 weighted avg      1.00      1.00      1.00        27

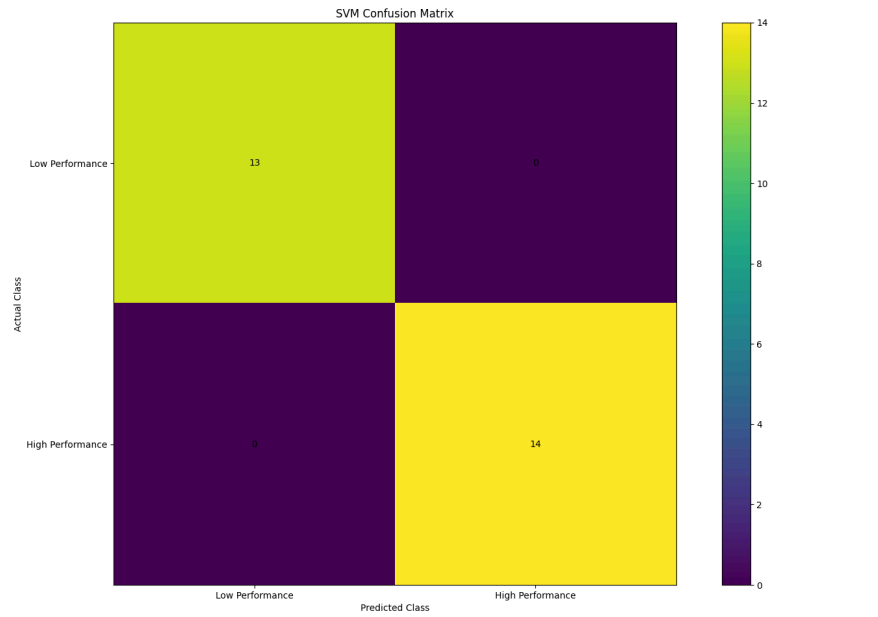
Confusion Matrix:
[[13  0]
 [ 0 14]]
```

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Figure 1



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